✅ Section 1: Dataset Preprocessing

Checklist Item	Status	Notes
Download CICIDS2017 dataset	✅	All 8 daily CSV files are loaded
Load & merge CSVs into single DataFrame	✅	All files concatenated correctly
Drop `Infinity` values (replace with NaN first)	✅	Done for numeric cols only
Drop all NaN rows/columns	✅	`dropna()` applied
Normalize all features using `MinMaxScaler` (scale to [0,1])	❌ **MISSING**	`data_cleaning.ipynb` saves the raw cleaned CSV without scaling. `ids_modelling.ipynb` applies `StandardScaler`, **not** `MinMaxScaler`. The checklist explicitly requires `MinMaxScaler` — this is wrong and will cost marks
Encode labels as binary: BENIGN → 0, attacks → 1	✅	Correctly done
Handle class imbalance using SMOTE	❌ **MISSING**	Not present in either notebook. This is explicitly required and directly tied to rubric marks
Split dataset into train/test (80/20, stratified)	✅	Done in `ids_modelling.ipynb`
Save cleaned dataset as CSV	✅	Saved to Data/Cleaned Data/cleaned_data.csv

❌ Section 1 EDA (for paper figures)

Checklist Item	Status	Notes
Plot class distribution (before AND after SMOTE)	❌ **MISSING**	The raw class distribution is checked via `.value_counts()` but never plotted. There's no SMOTE, so no after-SMOTE plot either
Plot feature correlation heatmap	❌ **MISSING**	Not in either notebook
Print dataset shape, feature names, class counts	⚠️ Partial	Shape and dtypes are printed; feature names are printed after stripping; class counts shown via `.value_counts()` but not clearly formatted

✅ Section 2: Baseline Model (`ids_modelling.ipynb`)

Checklist Item	Status	Notes
Train RF using **all features** with default hyperparameters	✅	Done
Accuracy	✅	Recorded
Precision, Recall, F1	✅	Recorded
Detection Rate (True Positive Rate)	⚠️	`recall` is TPR, but it is not *labelled* as "Detection Rate" — risky if the marker checks explicitly
False Positive Rate (FPR)	✅	Calculated from confusion matrix
Number of features used (= 78)	❌ **MISSING**	The number of features is never printed or recorded for the baseline. Your dataset has 70 features after cleaning (not 78 — see below)
Training runtime	✅	Timed
Inference runtime	❌ **MISSING**	The time to run `rf.predict()` is never measured
Plot ROC curve for baseline model	❌ **MISSING**	Not present
Save baseline results to CSV	❌ **MISSING**	Not saved

❌ Section 3: Fitness Function

Checklist Item	Status	Notes
Implement the unified fitness function: `0.7×Acc + 0.2×(1−FPR) − 0.1×(features/total)`	❌ **WRONG**	The fitness function uses **F1 score only**. It completely ignores the required formula. This is explicitly tied to rubric marks
Function accepts a binary feature mask OR hyperparameter vector	⚠️	Accepts a combined individual (bits + hyperparams) rather than separately
Function trains RF on selected features/hyperparams and returns fitness	✅	Yes, RF is trained internally
Validate fitness function on known inputs	❌ **MISSING**	No test/validation of the fitness function

⚠️ Section 4: Genetic Algorithm

Checklist Item	Status	Notes
78-bit binary chromosome (1 = feature included)	❌	Hardcoded to `n_features` (70, not 78). This is because cleaning removed features; document this
Population size = 30	❌	`pop_size = 20` (checklist says 30)
Tournament or roulette wheel selection	✅	Tournament selection implemented
Single-point or uniform crossover (rate = 0.8)	⚠️	Single-point crossover implemented but **crossover rate of 0.8 is not applied** — every pair always crosses over
Bit-flip mutation (rate = 0.02)	❌	Mutation rate is `0.05`, not `0.02`
Run for 50 generations	❌	Only runs for 30 generations
Track best fitness per generation	✅	`best_fitness_history` tracked
Extract best chromosome → re-train RF on selected features	✅	Done
Record all evaluation metrics	✅	Accuracy, Precision, Recall, F1, FPR done
Save GA results & selected features to CSV	❌ **MISSING**	Not saved

❌ Section 7: Hybrid GA + PSO

Checklist Item	Status	Notes
Stage 1: GA finds optimal feature subset	❌ **MISSING**	Not implemented in either notebook
Stage 2: PSO tunes hyperparams on GA-selected features	❌ **MISSING**	Not implemented
Record metrics for combined pipeline	❌ **MISSING**	
Save Hybrid results to CSV	❌ **MISSING**	

❌ Section 8: Evaluation & Visualizations

Checklist Item	Status	Notes
Comparison DataFrame (Baseline, GA, PSO, SA, Hybrid)	❌	Left as `# TODO` in Cell 25
Convergence curves for each metaheuristic	❌	Left as `# TODO` in Cell 27. History arrays exist but plots are not made
ROC curves — all methods on one graph	❌	Left as `# TODO` — completely missing
Bar chart — features selected per method	❌	Left as `# TODO` in Cell 26
Pareto front (optional)	❌	Not done
Feature importance — which features GA selected most	❌	Not done
Save plots as PNG files	❌	Not done
Save final comparison table as CSV	❌	Not done

⚠️ Section 9: Code Quality

Checklist Item	Status	Notes
Modular scripts (preprocessing.py, baseline.py, etc.)	❌	Everything is in notebooks; no `.py` files
Docstrings on all functions	⚠️	Only `get_neighbour()` and `decode_particle()` have docstrings; `fitness()`, `tournament_selection()`, `crossover()`, `mutate()`, `create_individual()` have **none**
Consistent snake_case naming	✅	Generally followed
README.md with setup + run instructions	❌	Not present
requirements.txt	❌	Not present
Test scripts run end-to-end	❌	Multiple TODO cells would cause errors
Submission ZIP	❌	Not prepared

🔴 Priority Summary — Critical Gaps to Fix First

Priority	Issue
🔴 **Critical**	SMOTE is completely missing — class imbalance not handled
🔴 **Critical**	Fitness function uses F1 only, not the required `0.7×Acc + 0.2×(1−FPR) − 0.1×(features/total)` formula
🔴 **Critical**	Hybrid GA+PSO is completely unimplemented
🔴 **Critical**	All Section 8 evaluation plots/tables are `# TODO` stubs
🔴 **Critical**	`MinMaxScaler` required but `StandardScaler` used instead
🟠 **High**	GA parameters wrong: pop=20 (should be 30), generations=30 (should be 50), mutation rate=0.05 (should be 0.02), crossover rate not enforced
🟠 **High**	PSO only tunes 2 hyperparams; `min_samples_split` and `min_samples_leaf` missing
🟠 **High**	No results are saved to CSV (Baseline, GA, PSO, SA)
🟠 **High**	No ROC curve for any method
🟡 **Medium**	EDA plots (class distribution, correlation heatmap) missing
🟡 **Medium**	Inference runtime not measured for baseline
🟡 **Medium**	No docstrings on most functions
🟡 **Medium**	No README.md or requirements.txt